

Longitudinal Vehicle Dynamics: Point-Mass Model, State Variables, Inputs, and ODE Solution

1 Longitudinal Vehicle Dynamics

Longitudinal dynamics describes the motion of a vehicle along its longitudinal axis, i.e. its forward and backward motion.

For a simple point-mass vehicle model, the complete vehicle is represented by a single mass m moving along a straight line. Rotational, lateral, suspension, and individual wheel dynamics are neglected.

The vehicle position is described by the longitudinal coordinate $s(t)$. The corresponding velocity and acceleration are

$$v(t) = \dot{s}(t), \quad (1)$$

and

$$a(t) = \dot{v}(t) = \ddot{s}(t). \quad (2)$$

Thus,

Symbol	Meaning	Unit
s	longitudinal position	m
v	longitudinal velocity	m s^{-1}
a	longitudinal acceleration	m s^{-2}
m	vehicle mass	kg

The point-mass assumption means that the vehicle is treated as a single translational mass. There is no yaw angle, yaw rate, sideslip angle, suspension motion, or individual wheel dynamics in this model.

2 Force Balance

The longitudinal equation of motion follows directly from Newton's second law,

$$\sum F_x = ma. \quad (3)$$

For a flat road, the longitudinal forces considered in this model are:

- driving or tractive force F_{drive} ,
- aerodynamic drag F_{drag} ,
- rolling resistance F_{roll} .

Therefore,

$$\boxed{ma = F_{\text{drive}} - F_{\text{drag}} - F_{\text{roll}}} \quad (4)$$

or equivalently,

$$\boxed{F_{\text{drive}} = F_{\text{drag}} + F_{\text{roll}} + ma.} \quad (5)$$

The acceleration is therefore

$$\boxed{a = \frac{F_{\text{drive}} - F_{\text{drag}} - F_{\text{roll}}}{m}.} \quad (6)$$

3 Longitudinal Forces

3.1 Driving Force

The driving force is denoted by

$$F_{\text{drive}}. \quad (7)$$

In the present model, it is treated as a constant input:

$$F_{\text{drive}} = F_0. \quad (8)$$

In a more detailed vehicle model, the driving force could depend on vehicle speed, throttle position, gear ratio, engine torque, motor torque, or a powertrain map. None of these details are required for the present point-mass model.

3.2 Aerodynamic Drag

Aerodynamic drag is modelled as

$$\boxed{F_{\text{drag}} = \frac{1}{2}\rho C_D A v^2} \quad (9)$$

where

$$\rho = \text{air density}, \quad (10)$$

$$C_D = \text{aerodynamic drag coefficient}, \quad (11)$$

$$A = \text{frontal area}. \quad (12)$$

The important property is the quadratic dependence on velocity,

$$F_{\text{drag}} \propto v^2. \quad (13)$$

Therefore, aerodynamic drag becomes increasingly important as vehicle speed increases.

3.3 Rolling Resistance

A simple constant rolling-resistance model is

$$\boxed{F_{\text{roll}} = C_{rr}mg} \quad (14)$$

where

$$C_{rr} = \text{coefficient of rolling resistance}, \quad (15)$$

$$g = \text{gravitational acceleration}. \quad (16)$$

The coefficient C_{rr} is dimensionless. It represents the resistance associated primarily with tire deformation and the tire–road interaction. It is not a longitudinal slip coefficient.

For a typical passenger-car model, a value such as

$$C_{rr} = 0.01 \quad (17)$$

is reasonable for an introductory model.

4 Complete Longitudinal Equation

Substituting the aerodynamic drag and rolling resistance equations into Newton’s second law gives

$$ma = F_{\text{drive}} - \frac{1}{2}\rho C_D A v^2 - C_{rr}mg. \quad (18)$$

Therefore,

$$\boxed{\dot{v} = \frac{F_{\text{drive}} - \frac{1}{2}\rho C_D A v^2 - C_{rr}mg}{m}}. \quad (19)$$

Together with the kinematic relation

$$\boxed{\dot{s} = v}, \quad (20)$$

these two equations completely define the point-mass longitudinal model.

5 State Variables

To express the model as a first-order ODE, define the state vector as

$$\boxed{\mathbf{x} = \begin{bmatrix} v \\ s \end{bmatrix}}. \quad (21)$$

The state derivative is

$$\boxed{\dot{\mathbf{x}} = \begin{bmatrix} \dot{v} \\ \dot{s} \end{bmatrix} = \begin{bmatrix} a \\ v \end{bmatrix}}. \quad (22)$$

The two states have different roles:

- v describes the current longitudinal velocity.
- s describes the current longitudinal position.

The position s is a state because it is obtained by integrating velocity:

$$s(t) = s(t_0) + \int_{t_0}^t v(\tau) d\tau. \quad (23)$$

Thus, even though s does not directly affect the forces in this simple model, it is included as a state when the vehicle position is required as an output.

6 Input

The driving force can be defined as the model input,

$$\boxed{u = F_{\text{drive}}.} \quad (24)$$

With this notation, the longitudinal model becomes

$$\dot{v} = \frac{u}{m} - \frac{\rho C_D A}{2m} v^2 - C_{rr} g, \quad (25)$$

and

$$\dot{s} = v. \quad (26)$$

Therefore,

$$\boxed{\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u).} \quad (27)$$

The parameters of the model are

$$m, \quad C_{rr}, \quad C_D, \quad \rho, \quad A, \quad g. \quad (28)$$

For the present simulation, these parameters are treated as constants.

7 Nonlinear State-Space Form

The complete model can be written as

$$\dot{\mathbf{x}} = \begin{bmatrix} \frac{u - \frac{1}{2}\rho C_D A v^2 - C_{rr} m g}{m} \\ v \end{bmatrix}. \quad (29)$$

The model is nonlinear because of the term

$$v^2. \quad (30)$$

Consequently, the model cannot globally be written in the linear form

$$\dot{\mathbf{x}} = A\mathbf{x} + Bu \quad (31)$$

with constant matrices A and B .

Instead, it has the nonlinear form

$$\boxed{\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u).} \quad (32)$$

8 Why $\dot{\mathbf{x}}$ Is Calculated

The ODE solver requires the time derivative of the state vector.

For

$$\mathbf{x} = \begin{bmatrix} v \\ s \end{bmatrix}, \quad (33)$$

the derivative is

$$\dot{\mathbf{x}} = \begin{bmatrix} \dot{v} \\ \dot{s} \end{bmatrix} = \begin{bmatrix} a \\ v \end{bmatrix}. \quad (34)$$

The derivative function therefore does not itself perform the numerical integration. It provides the instantaneous rate of change of the states to the ODE solver.

The computational chain is

$$\boxed{F_{\text{drive}} \rightarrow F_{\text{net}} \rightarrow a \rightarrow v \rightarrow s.} \quad (35)$$

More explicitly,

$$F_{\text{net}} = F_{\text{drive}} - F_{\text{drag}} - F_{\text{roll}}, \quad (36)$$

then

$$a = \frac{F_{\text{net}}}{m}, \quad (37)$$

then

$$\dot{v} = a, \quad \dot{s} = v. \quad (38)$$

9 Numerical Integration with ode45

MATLAB's `ode45` solves the first-order ODE numerically.

The derivative function can be written as

```
function xdot = longitudinal_dynamics(t, x, F_drive, m, Crr, Cd, rho, A)

    v = x(1);

    F_drag = 0.5 * rho * Cd * A * v^2;
    F_roll = Crr * m * 9.81;

    a = (F_drive - F_drag - F_roll) / m;

    xdot = [a; v];

end
```

The function receives the current state

$$\mathbf{x} = \begin{bmatrix} v \\ s \end{bmatrix} \quad (39)$$

and returns

$$\dot{\mathbf{x}} = \begin{bmatrix} a \\ v \end{bmatrix}. \quad (40)$$

The actual numerical integration is performed by `ode45`, for example:

```
[t,X] = ode45(@(t,x) ...
    longitudinal_dynamics(t, x, F_drive, m, Crr, Cd, rho, A), ...
    t_span, x0, opts);
```

The relationship is therefore

$$\boxed{\text{ode45} \rightarrow \text{longitudinal_dynamics} \rightarrow \dot{\mathbf{x}} \rightarrow \text{numerical integration.}} \quad (41)$$

The function `longitudinal_dynamics` does not call `ode45`; rather, `ode45` repeatedly calls the derivative function.

10 Initial Conditions

The initial state is specified as

$$\mathbf{x}_0 = \begin{bmatrix} v_0 \\ s_0 \end{bmatrix}. \quad (42)$$

For a vehicle starting from rest at the reference position,

$$\boxed{\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.} \quad (43)$$

This means

$$v(0) = 0, \quad s(0) = 0. \quad (44)$$

The simulation time interval can be specified as

$$t \in [0, 60] \text{ s.} \quad (45)$$

11 Analytical Validation Without Aerodynamic Drag

A useful validation case is obtained by setting

$$C_D = 0. \quad (46)$$

The drag force then becomes

$$F_{\text{drag}} = 0. \quad (47)$$

The acceleration is constant:

$$a = \frac{F_{\text{drive}} - C_{rr}mg}{m}. \quad (48)$$

For zero initial velocity and position,

$$v(0) = 0, \quad s(0) = 0, \quad (49)$$

the analytical solutions are

$$\boxed{v(t) = at} \quad (50)$$

and

$$\boxed{s(t) = \frac{1}{2}at^2.} \quad (51)$$

These analytical solutions provide a direct validation of the numerical `ode45` implementation. The maximum numerical errors can be evaluated as

$$e_v = \max_t |v_{\text{ODE}}(t) - v_{\text{analytical}}(t)|, \quad (52)$$

and

$$e_s = \max_t |s_{\text{ODE}}(t) - s_{\text{analytical}}(t)|. \quad (53)$$

12 Terminal Velocity

For constant driving force, the vehicle approaches a terminal velocity when

$$a = 0. \quad (54)$$

Therefore,

$$F_{\text{drive}} = F_{\text{drag}} + F_{\text{roll}}. \quad (55)$$

Substituting the force models gives

$$F_{\text{drive}} = \frac{1}{2}\rho C_D A v_{\text{max}}^2 + C_{rr}mg. \quad (56)$$

Solving for terminal velocity gives

$$\boxed{v_{\text{max}} = \sqrt{\frac{2(F_{\text{drive}} - C_{rr}mg)}{\rho C_D A}}.} \quad (57)$$

This expression is valid when

$$F_{\text{drive}} > C_{rr}mg. \quad (58)$$

The terminal velocity illustrates the effect of aerodynamic drag: as velocity increases, the quadratic drag force increases until the total resistance balances the available driving force.

13 0–100 km/h Performance

The target velocity for a 0–100 km h^{−1} sprint is

$$v_{\text{target}} = \frac{100}{3.6} = 27.777 \dots \text{ m s}^{-1}. \quad (59)$$

The corresponding time can be obtained from the numerical solution by interpolation:

```
v_target = 100 / 3.6;
t_0_100 = interp1(v_base, t_base, v_target);
```

This determines the time at which the simulated velocity reaches 100 km h^{−1}.

14 Sensitivity to Aerodynamic Drag

The aerodynamic drag coefficient C_D directly affects the aerodynamic resistance,

$$F_{\text{drag}} = \frac{1}{2}\rho C_D A v^2. \quad (60)$$

For a fixed vehicle and driving force, increasing C_D increases the aerodynamic drag force at a given velocity.

The terminal velocity satisfies

$$v_{\text{max}} = \sqrt{\frac{2(F_{\text{drive}} - C_{rr}mg)}{\rho C_D A}}. \quad (61)$$

Therefore,

$$v_{\text{max}} \propto \frac{1}{\sqrt{C_D}}. \quad (62)$$

A sensitivity study can therefore evaluate different values of C_D , for example,

$$C_D \in \{0.25, 0.30, 0.35\}, \quad (63)$$

and compare their corresponding 0–100 km h^{−1} times and terminal velocities.

15 Example Model Parameters

The example simulation uses the following parameters:

Parameter	Value	Unit
m	1500	kg
C_{rr}	0.01	–
C_D	0.30	–
ρ	1.224	kg m ^{−3}
A	2	m ²
F_{drive}	3500	N
g	9.81	m s ^{−2}

These values are illustrative parameters for the simplified point-mass model.

16 Complete Model Summary

The complete flat-road point-mass longitudinal model is

$$\dot{v} = \frac{F_{\text{drive}} - \frac{1}{2}\rho C_D A v^2 - C_{rr}mg}{m} \quad (64)$$

and

$$\dot{s} = v. \quad (65)$$

With

$$\mathbf{x} = \begin{bmatrix} v \\ s \end{bmatrix}, \quad (66)$$

the model is

$$\dot{\mathbf{x}} = \begin{bmatrix} \frac{F_{\text{drive}} - \frac{1}{2}\rho C_D A v^2 - C_{rr} m g}{m} \\ v \end{bmatrix}. \quad (67)$$

The model therefore represents the complete longitudinal dynamics of the simplified point-mass vehicle under the assumptions of:

- straight-line motion,
- flat road,
- constant vehicle mass,
- constant rolling-resistance coefficient,
- constant aerodynamic parameters,
- constant air density,
- prescribed driving force,
- no explicit wheel rotational dynamics,
- no lateral or yaw dynamics.

17 MATLAB Simulation Results

The MATLAB implementation was executed using the point-mass longitudinal dynamics model described above. The zero-drag case was first used to validate the numerical integration against the analytical solution.

The maximum absolute errors obtained from the validation were

$$e_v = 2.8422 \times 10^{-14} \text{ m s}^{-1}, \quad (68)$$

and

$$e_s = 4.5475 \times 10^{-13} \text{ m}. \quad (69)$$

The numerical solution therefore agrees with the analytical solution to essentially machine precision.

The aerodynamic drag coefficient was then varied while keeping the other parameters fixed. The resulting 0–100 km h^{−1} sprint times and terminal velocities were:

C_D	0–100 km h ^{−1} Time [s]	v_{max} [km h ^{−1}]
0.25	12.73	376.8
0.30	12.80	344.0
0.35	12.86	318.5

The results show that increasing C_D increases aerodynamic resistance. Consequently, the simulated 0–100 km h^{−1} time increases slightly, while the terminal velocity decreases.

The MATLAB command-window output was:

```
>> longitudinal_dyn
Validation -> Velocity Error: 2.8422e-14 m/s | Displacement Error: 4.5475e-13 m

Cd | 0-100 Time (s) | V_max (km/h)
0.25 | 12.73 | 376.8
0.30 | 12.80 | 344.0
0.35 | 12.86 | 318.5
>>
```